

LLM Abstract Screening for CONSORT Eligibility: A Solve-or-Guess Validation

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Abstract

Large language models are increasingly used to screen abstracts, deciding whether a study meets a set of eligibility criteria. Validating such a screener is hard when the criterion has genuine edge cases where conscientious reviewers may disagree, so a single human answer key cannot separate a screener’s mistakes from ordinary disagreement, and a key built by the same reviewers makes the comparison circular. We validate an LLM screener without a gold standard using the Solve or Guess model, which treats the LLM screener as one rater among several other human raters and estimates how often each rater solves the task from agreement alone. We demonstrate the method on a screening task of practical consequence: identifying the individually randomized trials of patient level health outcomes to which the CONSORT checklist applies, which will be used as the eligibility determination for our reporting quality trial at *The BMJ*. Three human reviewers and one LLM labeled 100 recent medRxiv abstracts, drawn to concentrate hard cases. The LLM’s estimated solving probability is consistent with all three reviewers: every kappa ratio interval against a human contains one, and its pairwise agreement matches the humans’ agreement among themselves. With 100 abstracts the design cannot certify equivalence within a tight margin, but it detects no gap between the LLM and the human reviewers, established without a gold standard. The method is generalizable. Any review can validate its own LLM screener the same way: have two or more human reviewers label the same abstracts, then compare their solving probabilities without relying on a gold standard.

1 Introduction

Large language models are increasingly trusted to make judgment calls that people used to make by hand. Abstract screening is one of them. To decide

whether a study belongs to a review or an audit, someone reads its title and abstract and judges it against a set of eligibility criteria, and an LLM can now render that judgment in seconds. Before we let an automated screener make the call, we want to know whether it makes the call as well as a person does. That question is easy when a trusted answer key exists. This paper takes on the harder case, where no such key can be relied on.

The nearest body of work automates screening for systematic reviews. Classical machine learning treated it as semiautomation: an active learner reranks the unscreened abstracts as a reviewer labels, surfacing likely includes first so the reviewer can stop early while still catching most of them (Chai et al., 2021). Large language models pushed toward full automation, used directly as zero shot classifiers, with ensembles reaching high sensitivity and specificity on curated benchmarks such as the Cochrane reviews (Sanghera et al., 2024) and neural rerankers ranking abstracts against the inclusion criteria (Jaumann et al., 2025). These systems differ in design but are judged the same way, against a human curated gold standard on sensitivity, specificity, and F1 (Kim et al., 2025; Sanghera et al., 2024), with prompts tuned toward near complete recall because a review cannot afford to miss relevant studies. The scoring strains in the low prevalence, cost asymmetric setting where screening actually happens, and recent guidance warns that standard confusion matrix metrics mislead there (Madeyski et al., 2025).

Every one of these frameworks treats the human benchmark as infallible. When the screener and a reviewer disagree, the metrics count it as a screener error, even though a single human key cannot tell an LLM mistake from ordinary disagreement between conscientious reviewers. In practice the key is often built by the same reviewers the screener is later scored against, which makes the comparison circular. For an eligibility criterion with genuine edge cases, where experienced reviewers themselves disagree, this is the normal situation rather than a corner case, and the gold standard the comparison rests on is exactly what is in doubt.

We validate the screener with the Solve or Guess model of Rohe et al. (2026). The model treats the automated screener as one rater among several and asks how often each rater solves the task, rather than how often it matches a designated key. Each rater a has a solving probability p_a . Under the model, Cohen’s κ between two raters factors into the product of their solving probabilities, so the ratio of two raters’ agreement with a common third rater recovers the ratio of their solving probabilities. The comparison needs no gold standard, and it applies to any screening decision that several raters make on the same items. If the screener agrees with each human about as often as the humans agree with one another, it solves the task

about as well as they do.

We put this to work on a practical task: identifying manuscripts for a CONSORT reporting checklist compliance study. The individually randomized trials that evaluate patient level health outcomes, the studies to which the CONSORT reporting checklist applies, are eligible for our study. CONSORT specifies the minimum information a randomized trial report should contain, such as how randomization was done and who was blinded; it is endorsed by hundreds of journals, and its current version is a 30-item checklist updated in 2025 (Hopewell et al., 2025). This eligibility decision determines which submissions enter our randomized trial of an LLM CONSORT reporting intervention at *The BMJ* (Krauska et al., 2026). Section 2 describes that trial and why we built the corpus to stress the decision rather than to sample it evenly.

Three human reviewers and one LLM labeled the same 100 recent medRxiv abstracts as Include or Exclude, the LLM working from a codebook prompt at a fixed temperature of 0. We fit the Solve or Guess model to the four raters. The LLM’s estimated solving probability is consistent with the three human reviewers’: every kappa ratio interval against a human contains one, and its pairwise agreement with the humans matches theirs with one another. With 100 abstracts, the design cannot certify equivalence within a tight margin, so we do not claim the LLM is exactly as reliable as a person. We find no detectable gap between them, and we establish this without a gold standard.

This paper provides a repeatable methodology. Solve or Guess certifies an LLM screener against several human reviewers without a gold standard, and CONSORT eligibility screening is the worked example. The accompanying code repo is reproducible end to end, so another review can rerun the validation on its own eligibility criteria and its own reviewers. Section 2 states the purpose and the corpus design. Section 3 states the task, the codebook, and the model. Section 4 reports the estimates. Section 5 discusses what the finding does and does not state, for this task and for screening more generally.

2 The worked example: our CONSORT reporting trial

We are running a randomized trial of an LLM-generated intervention to improve CONSORT reporting at *The BMJ* (Krauska et al., 2026). The journal receives far more submissions than the intervention can act on, as only a

fraction are the individually randomized trials to which CONSORT applies. Each submission must be evaluated for our study’s eligibility criteria: is this an individually randomized trial that evaluates patient level health outcomes, the kind of study that should be checked against CONSORT? We will use an LLM to make this eligibility determination in the trial. This paper is the validation for this LLM eligibility determination.

This validation study intentionally selects for the difficult cases. Most submissions are easy. A paper with no connection to randomized trials is excluded correctly by any reader, and it reveals nothing about whether the LLM can be trusted. The decision that matters is the hard one, where a manuscript uses words related to randomization yet still fails the criterion. To concentrate these cases, we drew the corpus from medRxiv preprints returned by a search for the word “random.” The choice deliberately raises the prevalence of papers that could be mislabeled. Cluster randomized trials, random effects meta-analyses, Mendelian randomization studies, randomly sampled observational cohorts, and random forest models all use the word yet fall outside the target population. Screening this corpus is therefore a demanding test. The raters must separate genuine individually randomized outcome trials from the many near misses that share their vocabulary, rather than sort obvious includes from obvious excludes.

A screener that performs on the same level as human reviewers across this enriched corpus has therefore been tested where screening is most likely to fail. This search does leave one stratum unexamined, the eligible trials that never use the word “random,” and we take up that gap and its consequences for deployment in Section 5.

3 Methods

We estimate how often an LLM solves the abstract screening task relative to human reviewers, on the same abstracts and without a gold standard label set. The estimate comes from the Solve or Guess model (Rohe et al., 2026), which summarizes each rater by a single solving probability and recovers those probabilities from agreement alone. This section states the screening task, describes the corpus and the raters, recalls the part of the model we use, and gives the estimation procedure. We rely on Rohe et al. (2026) for the model’s development and proofs, and restate only what the reader needs to follow the analysis.

3.1 The screening task

The task is to decide, from a title and abstract, whether a paper belongs to the target population for full text CONSORT review: individually randomized trials that evaluate patient level health outcomes. Each abstract receives one label, Include or Exclude.

3.2 The corpus and raters

We drew the evaluation corpus from medRxiv: the 100 most recent preprints whose title or abstract contains the word “random,” retrieved on 1 May 2026. Their posting dates range from 11 April to 30 April 2026. Section 2 explains why we built the corpus this way.

Four raters labeled the same 100 abstracts: three human reviewers and one LLM. Every rater labeled every abstract (a *complete design*), so all $\binom{4}{2} = 6$ rater pairs are observed on all 100 items. The three human reviewers worked from the same codebook as the LLM (Section 3.3), though they were not asked to record its chain of thought reasoning; each recorded only the final Include or Exclude label. The LLM ran the codebook as a single prompt with Claude Haiku 4.5 at temperature 0, and returned Human Review on 2 of the 100 abstracts. We collapse Human Review to Include, the conservative choice for a screening step that feeds a downstream full text audit.¹ The label space is then binary for all four raters.

The corpus also postdates the model’s training data, which ends in July 2025; the abstracts were posted in April 2026.² The LLM therefore could not have seen these papers during training, which rules out data contamination: its agreement with the human reviewers cannot be an artifact of having encountered the abstracts before. This is a built in strength of validating a screener on newly posted preprints.

3.3 The codebook

Careful codebook creation is an important implementation step. The comparison rests on solver consensus (Section 3.4): the assumption that each abstract has a single correct label that conscientious raters converge on when they solve. That assumption holds only as long as the codebook is sharp. A vague criterion lets two careful reviewers reach different labels for good

¹With only 2 of 100 abstracts affected, the alternative policies (collapse to Exclude, treat Human Review as a third label class, or drop the item) move the estimates negligibly.

²Claude Haiku 4.5 has a July 2025 training data cutoff, per Anthropic’s published model documentation.

reasons, and then there is no single target to recover, so the kappa ratio no longer measures solving ability. Fixing the eligibility question in six concrete steps is what makes solver consensus plausible here. The codebook is therefore a precondition for the whole comparison, and we treat its design as part of the method.

The LLM screens each abstract by working through a codebook, the prompt that operationalizes the eligibility criteria. The codebook poses six questions in order, which structure the model’s chain of thought. The first four collect evidence, the fifth synthesizes it, and the sixth records the decision:

1. whether the manuscript actively analyzes empirical data;
2. whether that analysis uses individual patient level data to evaluate health outcomes;
3. what allocation language the abstract contains, copied or paraphrased from the text;
4. whether that language indicates the intervention was individually randomized;
5. a step by step synthesis of the evidence from the first four questions;
6. the final decision: Include, Exclude, or Human Review, the last reserved for abstracts too ambiguous to call.

The codebook text is `prompts/screen.v1_codebook.txt`.

We wrote the codebook on a development set, deliberately held apart from the abstracts we evaluate. The development set is the 20 preprints that met the same “random” search criterion and were posted immediately before the evaluation corpus. We drafted and revised the six questions against these 20 abstracts, then froze the codebook before running it on the 100 evaluation abstracts. The two sets are disjoint, so the prompt was never tuned on the abstracts it is later scored against. We hold the codebook fixed for every abstract in the evaluation.

3.4 The Solve or Guess model

The Solve or Guess model (Rohe et al., 2026) describes how each rater produces each label. Each abstract i has a true label $Y_i \in \{\text{Include}, \text{Exclude}\}$, its true CONSORT eligibility status. When rater a reads abstract i , one of two things happens. With probability p_a , the rater solves the abstract:

they work through the eligibility question and recover the true label, so $R_{a,i} = Y_i$. Otherwise, the rater guesses: they draw a label from their own guessing distribution $\pi^{(a)}$, which does not depend on Y_i . The solving probability $p_a \in [0, 1]$ summarizes rater a in a single number on a scale shared by the humans and the LLM; it is the quantity we compare.

Two raters can agree because both solved and thus recovered Y_i , or because at least one guessed and the labels happened to match. The model separates the two. In the population, Cohen’s κ (Cohen, 1960) between two raters factors into the product of their solving probabilities, $\kappa(a, b) = p_a p_b$ (Rohe et al., 2026).³ This is a statement about the population κ , the value the sample statistic $\hat{\kappa}$ converges to, not about any single dataset’s observed $\hat{\kappa}$. The factorization has a corollary we use directly. For three raters a , b , and c , the ratio $\kappa(a, c)/\kappa(b, c) = p_a/p_b$ recovers the ratio of two raters’ solving probabilities, free of the reference rater c and free of any gold standard label. We read $p_{\text{LLM}}/p_{\text{human}} > 1$ as the LLM solving the task more often than that human. The comparison is robust to biased guessing: it survives even when the raters’ guessing distributions pull the individual κ values away from that clean product form (Rohe et al., 2026). Identification needs at least three raters, and our four rater design clears that minimum by one (Rohe et al., 2026). We estimate the solving probabilities by maximum likelihood with the EM algorithm described in the next section.

Two assumptions about our task let the agreement structure identify the solving probabilities. The first is *solver consensus*: each abstract admits a single Y_i for solvers to converge on, which is a statement that the codebook is sharp enough that conscientious reviewers would agree on the label. The second is *single source dependence*: each rater’s solving events and guesses are independent across raters and independent of Y_i , so the true label is the only source of dependence between raters. Agreement beyond chance therefore points back to Y_i rather than to shared bias or copying, and that is what makes the agreement structure informative about who is solving. The human reviewers labeled independently, and the LLM had no access to their decisions, so single source dependence holds by construction of the study.

The model is a constrained Dawid–Skene latent class model (Dawid and Skene, 1979). Dawid and Skene give each rater a full confusion matrix; Solve or Guess collapses that matrix to the single scalar p_a , and it is this constraint that yields the factorization above and the one number per rater

³The clean product form is exact when raters are unbiased guessers, drawing their guesses in proportion to the label frequencies τ . Rohe et al. (2026) give a general factorization that drops this assumption, and the EM estimator we use (Section 3.5) does not rely on it.

comparison.

3.5 Estimation and inference

We fit the homogeneous model, in which each rater’s solving probability is constant across abstracts. We estimate the parameters $(\tau, \{\pi^{(a)}\}, \{p_a\})$ by maximum likelihood with the EM algorithm of Rohe et al. (2026), treating the LLM as one rater and each human as a separate rater. Each rater receives its own guessing distribution, so the fit does not assume unbiased guessing.

We report three quantities:

1. the estimated solving probability $\hat{p}_a$ for each of the four raters, which orders them on a common scale;
2. the kappa ratio $\hat{p}_{\text{LLM}}/\hat{p}_{\text{human}}$ against each of the three humans, the primary comparison, where an interval that contains 1 means the data do not detect a difference between the LLM and that human;
3. the observed pairwise Cohen’s κ for all six rater pairs, a descriptive check on the raw agreement.

We quantify uncertainty with the nonparametric bootstrap of Rohe et al. (2026), resampling abstracts rather than individual labels so that the agreement structure within each abstract is preserved. We draw $n = 100$ abstracts with replacement, refit the model, and repeat $B = 1000$ times. For each solving probability and each kappa ratio we report the 95% bootstrap interval.

We check the model’s fit with the rank one diagnostic of Rohe et al. (2026). The idea is easy to state. If the model holds, one number per rater has to explain the entire pattern of pairwise agreements: once the chance term is removed, every pair’s excess agreement is the product $p_a p_b$. The diagnostic asks whether the observed pattern needs more than that, and it probes the two structural assumptions directly. Excess agreement concentrated on a single rater pair signals a shared source of agreement beyond the true label, such as the LLM and a human keying on the same surface cue, which is the violation that would inflate the kappa ratios reported here. A second, independent dimension of agreement signals raters splitting into camps that each solve toward a different answer, a failure of solver consensus. Formally, the off-diagonal agreement matrix is rank one plus a chance constant, $A_{ab} = c + (1 - c) p_a p_b$. We measure the departure by the off-diagonal

residual sum of squares $T = \sum_{a < b} (A_{ab} - c - (1 - c)p_a p_b)^2$, and obtain its null distribution by a parametric bootstrap that simulates datasets from the fitted model, refits, and recomputes T ($B = 1000$). A large p -value means the observed agreement is consistent with the model.

The screening run and the analysis are scripted end to end (`code/03_run_screening.py`, `code/04_run_solve_or_guess.R`, and `code/05_rank_one_diagnostic.R`); 04 calls the estimator and the bootstrap from Rohe et al. (2026). Every reported number traces back to the experiment configuration `experiments/exp_001_codebook.yaml`.

4 Results

We fit the homogeneous Solve or Guess model to the four raters on the 100 abstracts. We label the three human reviewers A, B, and C, and write LLM for the automated rater. We first summarize the labels themselves, then read off three model quantities: each rater’s solving probability, the kappa ratio of the LLM against each human, and the observed pairwise agreement.

4.1 The labeled corpus

Includes are rare. Each rater labeled between 17 and 22 of the 100 abstracts Include (Table 1), so about four in five abstracts are Excludes for every rater. The four raters agreed unanimously on 84 of the 100 abstracts, 12 Include and 72 Exclude, and split on the remaining 16. This is the low prevalence, imbalanced setting the introduction flagged. Raw percent agreement runs high here precisely because most abstracts are easy excludes, which is why we read the raters through solving probabilities rather than raw accuracy.

Table 1: Each rater’s marginal Include count and rate across the 100 abstracts. The rates sit in a narrow band, and Includes are the minority class for every rater.

Rater	Include (n)	Include rate
A	20	0.20
B	22	0.22
C	17	0.17
LLM	21	0.21

4.2 Solving probabilities

Table 2 reports the estimated solving probability for each rater with its 95% bootstrap interval. The LLM’s estimate is 0.85, close to the two highest human estimates (0.81 and 0.83) and above the third (0.67). The four intervals overlap heavily. The LLM’s interval, $[0.69, 1.00]$, contains the point estimate of two of the humans, and each human’s interval contains the LLM’s. The point estimates order the raters, but the intervals are wide enough that little should be read into the order.

Table 2: Estimated solving probability $\hat{p}_a$ for each rater, with 95% non-parametric bootstrap intervals ($B = 1000$). The estimates are close and the intervals overlap heavily.

Rater	$\hat{p}_a$	95% CI
LLM	0.85	$[0.69, 1.00]$
A	0.81	$[0.61, 0.99]$
B	0.83	$[0.64, 0.98]$
C	0.67	$[0.46, 0.88]$

4.3 The LLM against each human

The kappa ratio compares two raters directly, without a gold standard (Rohe et al., 2026). Table 3 reports $\hat{p}_{\text{LLM}}/\hat{p}_{\text{human}}$ against each reviewer. All three point ratios are at or above 1: 1.05 against A, 1.02 against B, and 1.26 against C. Every interval contains 1, so the data do not detect a difference between the LLM and any human. The intervals are wide, though. Each one also contains ratios well below 1, so it is equally consistent with the LLM solving somewhat less often than a human; what the intervals rule out is a large difference in either direction, not a small one. The widest comparison is against reviewer C, where the ratio is 1.26 with interval $[0.88, 1.98]$.

4.4 Observed agreement

Table 4 reports observed pairwise Cohen’s κ for all six rater pairs, a descriptive check on the raw agreement behind the model fit. The values range from 0.61 to 0.79, the moderate to substantial band. The LLM’s agreement with the humans (0.61 to 0.79) falls in the same range as the humans’ agreement with one another (0.70 to 0.76). The LLM is not an outlier in the agreement structure. The raw agreement behind these kappas is high, 88 to 93 of 100

Table 3: Kappa ratio $\hat{p}_{\text{LLM}}/\hat{p}_{\text{human}}$ against each human reviewer, with 95% bootstrap intervals. An interval containing 1 means the data do not detect a difference between the two raters.

Comparison	Ratio	95% CI
LLM / A	1.05	[0.81, 1.40]
LLM / B	1.02	[0.77, 1.39]
LLM / C	1.26	[0.88, 1.98]

for every pair; chance correcting it for the rarity of Includes is what brings the values down into the moderate to substantial band.

Table 4: Observed pairwise Cohen’s κ for all six rater pairs.

	A	B	C	LLM
A	—	0.76	0.70	0.79
B		—	0.71	0.73
C			—	0.61
LLM				—

4.5 Model fit

The rank one diagnostic checks whether one solving probability per rater can explain the entire pattern of pairwise agreement, or whether some pair agrees more than the model allows (Section 3.5). A failure would flag a shared source of agreement beyond the true label, such as the LLM and a human keying on the same surface cue, or reviewers splitting into camps that solve toward different answers. The diagnostic does not reject the rank one structure (parametric bootstrap $p = 0.38$, $B = 1000$), so the agreement among the four raters is consistent with the Solve or Guess model. With four raters and 100 abstracts the test has limited power, so this is reassurance rather than strong confirmation.

4.6 Where the LLM stood alone

On 3 of the 100 abstracts the LLM disagreed with all three human reviewers, and all three are the kind of edge case the corpus was built to contain (Section 2). The LLM’s recorded reasoning is defensible on each.

On two, the LLM excluded a paper the humans included, and in both the disagreement is about what counts as a patient level health outcome. One is a within subject randomized psilocybin study in ten healthy volunteers whose outcome is resting state brain connectivity; the LLM excluded it as a mechanistic neuroimaging study rather than a clinical outcome trial. The other randomized 3,274 adults to lung screening invitation messages and measured intention to attend; the LLM excluded it as a behavioral intention experiment rather than a health outcome trial. Both readings are reasonable, and a reviewer could go either way.

On the third, the humans excluded a secondary analysis that pools data from randomized trials and observational cohorts to validate a histology derived prognostic signature. Here the LLM did not confidently include the paper. It returned Human Review, flagging the mixed randomized and observational design and the missing allocation procedure, and our conservative collapse of Human Review to Include (Section 3.2) is what turned that hedge into a lone Include. The abstract also carries “Randomized Trials” in its title, a surface cue a screener can overweight.

None of the three is a clear screening error. Two are genuine boundary calls on the eligibility criterion, and the third is an artifact of the label collapse policy on an abstract the LLM itself flagged for a human. This is the behavior the edge case corpus was designed to surface.

4.7 Summary

The data do not detect a difference between the LLM and any of the three human reviewers. Every kappa ratio interval against a human contains 1, and the LLM’s pairwise agreement with the humans falls in the same range as the humans’ agreement among themselves. The intervals are wide, so this is an absence of a detectable gap rather than a demonstration of equivalence: with 100 abstracts the design cannot certify that the LLM matches a human reviewer within a tight margin. What we can say is that, on this task and this corpus, the LLM’s solving probability is consistent with human level performance, and we establish this without a gold standard.

5 Discussion

The validation method generalizes beyond this task. To validate an automated screener when no gold standard exists, add at least two more independent raters, ideally two humans, and have them label the same items. Three raters is the minimum the model needs to identify each rater’s solving

probability (Section 3.4), and Solve or Guess then estimates how well the screener performs without a key. CONSORT eligibility screening is the case we worked through. On that task the LLM solves about as often as the human reviewers. This is a model based reliability claim, following from the Solve or Guess factorization under solver consensus and single source dependence, the condition that the true label is the only source of agreement between raters (Rohe et al., 2026). It is not a claim of gold standard accuracy. We never measured the LLM against a key; we measured how its agreement with three reviewers compares to their agreement with one another, and the data do not separate the two.

Whether an LLM can replace a human rater is a context dependent decision that weighs the risks. In the case of the randomized controlled trial of CONSORT labeling (Krauska et al., 2026), this evidence makes us comfortable doing eligibility screening for the trial with a human reviewer confirming eligibility at the point of enrollment. The numbers are also specific to this task and this corpus. The estimand is tied to one eligibility criterion and to a sample of recent medRxiv preprints, and it should not be carried to a different criterion or a different literature without rerunning the analysis.

The comparison rests on two assumptions, and a different task setting can break either one (Rohe et al., 2026). The first is solver consensus. It says that each abstract has one correct label, and that conscientious reviewers reach it when they solve. Solver consensus holds only when the codebook is sharp. A vague criterion lets two careful reviewers label the same abstract differently for good reasons, and then there is no single target to recover. When that happens, repair the codebook before grading any rater against it. The second is single source dependence. It says that raters agree because they recover the same label, not because they copy each other or share a bias. Here the LLM never saw the human labels, so it could not have copied them. However, if humans discussed their ratings while making them, their agreement is inflated even when neither solved the task. The rank one diagnostic checks for both failures. It does not reject the model on our data, but with only four raters, the check is weak.

This task serves as a worked example for others who wish to use Solve or Guess, and is deliberately simple: one frozen codebook, one model, one pass at temperature 0, and a complete response design. We did not compare prompt variants or estimate the variance of repeated runs, and for a validation we did not need to. Searching for an optimal prompt or model is a different task from validating a fixed one, and it would take far more labeled abstracts than a validation does. We fit the homogeneous model, with a constant solving probability per rater; we did not collect the per item dif-

ficuity features that the heterogeneous extension would use, and identifying them is future work. The corpus is 100 abstracts, so the bootstrap intervals are wide and only sizable differences are detectable. The three reviewers and the LLM worked from the same codebook, which is what makes solver consensus plausible. However, if the codebook itself were wrong and there was a systematic error in the criteria, it would distort the comparison and would lead all four raters into the same mistake and inflate their agreement. That is why we drafted the codebook on a separate development set and froze it before the evaluation (Section 3.3).

Future work will push the method into settings that strain these assumptions. Here the codebook is sharp, the reviewers worked independently, and the LLM never saw their labels. Some reviewer pools train together or confer, and an automated screener may share systematic biases with the humans it is judged against, both of which break single source dependence and inflate agreement. Two tools from Rohe et al. (2026) bear on this directly. The rank one diagnostic tests the assumptions each time the method is applied and flags where the basic model should not be trusted. The multicomcommunity generalization allows different consensus answers per item, related to the community of the rater. It fits systematic disagreement, where reviewers from different traditions each solve toward a different label instead of converging on one. Lighter extensions remain open too: comparing prompt variants by their effect on the kappa ratio, ordinal screening decisions, and other models and review topics.

Automating a judgment task always raises the same question: should the machine make this call, and how would we know it is safe to let it? For many of these tasks, human answers can vary, and so alignment between a human and LLM does not tell us about how reasonably an LLM varies compared to humans. Solve or Guess answers the question without a key, and while taking this variation into account. It validates an automated screener against several human raters and returns a defensible answer even when those raters disagree. Systematic review screening is the prime use case: every review begins by deciding which studies are in scope, and any review whose reviewers label the same abstracts can certify its screener this way. We demonstrated the Solve or Guess algorithm on eligibility screening for a CONSORT study, where the LLM and the human reviewers show no detectable difference. We now use this screening in the trial itself, where it identifies the submissions eligible for CONSORT review at *The BMJ* (Krauska et al., 2026), and the papers it selects feed the downstream reporting quality work.

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